DEV BHOOMI UTTARAKHAND UNIVERSITY

project report on C++ QUIZ GAME

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INTRODUCTION

The **C++ Quiz Application** is a comprehensive console-based software project designed to evaluate technical proficiency and knowledge in computer science. In the modern educational era, digital assessments have become a primary tool for measuring learning outcomes. This application simulates a real-world testing environment where questions are presented sequentially and user responses are validated in real-time.

The project focuses on the core principles of procedural programming. It utilizes standard input and output streams to interact with the user, ensuring a seamless experience. The primary goal is to provide an interactive platform where students can test their knowledge while developers demonstrate control flow, state management, and algorithmic thinking.

By leveraging the robustness of C++, we have developed an engine that is both lightweight and efficient. This report details the technical architecture, implementation logic, and the systematic approach taken to build a reliable assessment platform.

PROJECT OBJECTIVES

The development of the C++ Quiz Engine was focused on achieving high standards in both software engineering and pedagogical assessment:

1. Technical Skill Enhancement: Deepening the understanding of C++ syntax, focusing on pre-processor directives, namespaces, and standard I/O management.

2. Robust Logic Implementation: Utilizing conditional control structures and Boolean logic to validate user data against pre-defined answer keys with high accuracy.

3. Dynamic User Feedback: Implementing a real-time scoring system that provides immediate performance metrics to the user upon completion of each task.

4. Professional Documentation: Adhering to the formal standards of technical reporting as required by academic and industry-level project submissions.

TECHNOLOGY USED

The following technical environment was established to facilitate the development and execution of the C++ Quiz Engine:

TECHNICAL CATEGORY	IMPLEMENTATION DETAILS
Programming Language	C++ (ISO Standard / C++11)
Development Environment	Visual Studio Code / CodeBlocks IDE
Compiler Infrastructure	GNU GCC Compiler Suite (MinGW)
Core Libraries	Standard Template Library (STL / iostream)
System Architecture	Console-Based Application Interface

SOURCE CODE IMPLEMENTATION

```
#include<iostream>
using namespace std;
class Question{public:string ques,op1,op2,op3,op4;int correct;
void setQuestion(string q,string a,string b,string c,string d,int
ans)

{ques=q;op1=a;op2=b;op3=c;op4=d;correct=ans;}
int askQuestion(){int choice;cout<<"\n"<<ques<<endl;
cout<<"1."<<op1<<"\n2."<<op2<<"\n3."<<op3<<"\n4."
<<op4<<endl;
cin>>choice;if(choice==correct){cout<<"Correct\n";return
1;}else{cout<<"Wrong\n";return 0;}}};

int main(){int score=0;Question q1,q2,q3,q4,q5,q6,q7,q8;
q1.setQuestion("Who developed
C++?","Dennis","Bjarne","James","George",2);
q2.setQuestion("Output operator?",">>","<<","==","=",2);
q3.setQuestion("File extension?",".c",".cpp",".txt",".java",2);
q4.setQuestion("Keyword for
class?","class","struct","define","obj",1);
q5.setQuestion("Input function?","cout","cin","print","scan",2);
q6.setQuestion("Loop with fixed
count?","for","while","do","switch",1);
q7.setQuestion("Decimal data
type?","int","char","float","bool",3);
q8.setQuestion("One name many
forms?","inherit","encap","poly","abs",3);
score+=q1.askQuestion();score+=q2.askQuestion();score+=q3.a
skQuestion();score+=q4.askQuestion();
score+=q5.askQuestion();score+=q6.askQuestion();score+=q7.a
skQuestion();score+=q8.askQuestion();
cout<<"\nFinal Score = "<<score<<"/8"<<endl;return 0;}
```

CONCLUSION & FUTURE SCOPE

In conclusion, the development of the C++ Quiz Application has been a successful exercise in applying theoretical programming concepts to a practical software solution. The application effectively handles user inputs, validates logic through conditional structures, and provides an accurate scoring system.

The project demonstrates that C++ is an exceptionally powerful tool for developing educational software. By utilizing structured programming techniques, we have created a reliable engine that represents the foundational building blocks of modern software engineering.

FUTURE ROADMAP

- **GUI Integration:** Migration from a console-based UI to a Graphical User Interface for better accessibility.
- **Database Persistence:** Integrating SQL-based storage to maintain a history of user performance.
- **Dynamic Assessments:** Adding a timer-based competitive mode to enhance user engagement.

REFERENCE

The development of this C++ Quiz Game project was done with the help of various learning resources.
Basic concepts of C++ were understood from standard textbooks like :
The C++ Programming Language by Bjarne Stroustrup
Object Oriented Programming with C++ by E. Balagurusamy.
Online platforms such as GeeksforGeeks and W3Schools
Tutorialspoint were also used to learn syntax, improve logic building, and understand program structure.
In addition, classroom notes and teacher guidance helped in completing the project successfully.

THANKYOU!